Tutorial - 1

1.4 Practice Problem on Preliminary Concepts:

1. Given Sets

$A = \{x : x \in N \text{ and } x \text{ is even}\}$

So, $A = \{2, 4, 6, 8, 10, \ldots\}$

$B = \{x : x \in N \text{ and } x \text{ is prime}\}$

So, $B = \{2, 3, 5, 7, 11, \ldots\}$

$C = \{x : x \in N \text{ and } x \text{ is a multiple of } 5\}$

So, $C = \{5, 10, 15, 20, \ldots\}$

a) $A \cap B$ (The intersection of sets $A \cap B$):

$= \{2\}$

b) $B \cap C$ (The intersection of sets B and C):

$= \{5\}$

c) $A \cup B$ (The union of sets A and B):

$= A \cup B = \{2, 3, 4, 5, 6, 7, 8, 10, 11, \ldots\}$

d) $A \cap (B \cup C)$ (The intersection of set A and the union of set B and C):

$= B \cup C = \{2, 3, 5, 7, 10, 11, 15, 20, \ldots\}$

$A \cap (B \cup C) = \{2, 10\}$

2. Prove:

i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Let x be an element.

If $x \in A \cup (B \cap C)$

Then $x \in A$ or $x \in (B \cap C)$

If $x \in (B \cap C)$

Then $x \in B$ and $x \in C$

So $x \in A$ or ($x \in B$ and $x \in C$).

This implies

($x \in A$ or $x \in B$) and ($x \in A$ or $x \in C$)

$\therefore x \in (A \cup B) \cap (A \cup C)$

Hence proved:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Let x be an element

If $x \in A \cap (B \cup C)$

Then $x \in A$ and $x \in (B \cup C)$

If $x \in (B \cup C)$

Then $x \in B$ or $x \in C$

So, $x \in A$ and $(x \in B$ or $x \in C)$

This implies

$(x \in A$ and $x \in B)$ or $(x \in A$ and $x \in C)$

$\therefore x \in (A \cap B) \cup (A \cap C)$

Henced proved:

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

2. Matrix Operation and System of Linear Equations.

2.1. Practice problem on Basic Matrix Operation

1.a.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b.
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

c.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Given:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$AD = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 6 & 12 \\ 2 & 9 & 16 \end{bmatrix}$$

$$DA = \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 4 & 12 & 16 \end{bmatrix}$$

$AD \neq DA$.

So matrix multiplication is not commutative.

3. Given:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$$AC = \begin{bmatrix} 4 & 7 \\ 8 & 14 \end{bmatrix}$$

$\therefore AB = AC$.

4. Given:

A is an $m \times n$ matrix

B is an $n \times p$ matrix

We want to prove:

$$(AB)^T = B^T A^T$$

Step 1: Definition of transpose.

If $C = [c_{ij}]$, then $C^T = [c_{ji}]$

Rows become columns.

Step 2: Compute element of $(AB)^T$

Let $AB = C$, then

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

So, the (i,j)-th element of the transpose is

$$(AB)^T_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

Step 3: Compute element of $B^T A^T$

$$(B^T A^T)_{ij} = \sum_{k=1}^{n} (B^T)_{ik} (A^T)_{kj} = \sum_{k=1}^{n} b_{ki} a_{jk}$$

Step 4: Compare.

$$(AB)^T_{ij} = \sum_{k=1}^{n} a_{jk} b_{ki} = \sum_{k=1}^{n} b_{ki} a_{jk} = (B^T A^T)_{ij}$$

$$\therefore (AB)^T = B^T A^T$$

5. Given:

$$A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$$

$$A^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

$$A^T A = \begin{bmatrix} a^2+b^2+c^2 & ad+be+cf \\ ad+be+cf & d^2+e^2+f^2 \end{bmatrix}$$

$A^T A$ is symmetric because the off-diagonal terms are equal.

Sum of diagonal elements:

$$(a^2+b^2+c^2)+(d^2+e^2+f^2)$$
$$= a^2+b^2+c^2+d^2+e^2+f^2$$

This equals the sum of squares of all elements of A.

6. For a 2x2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then,

$$A^{-1} = \left(\frac{1}{(ad-bc)}\right) * \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Matrix A:

$$A = \begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$$

$$\det(A) = (5\times5 - 6\times4)$$
$$= 25 - 24$$
$$= 1$$

$$A^{-1} = \left(\frac{1}{1}\right) \times \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$
$$= \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix}$$

Matrix B:

$$B = \begin{bmatrix} 6 & 4 \\ 3 & 3 \end{bmatrix}$$

$$\det(B) = (6\times3 - 4\times3) = 18 - 12 = 6$$

$$B^{-1} = \left(\frac{1}{6}\right) \times \begin{bmatrix} 3 & -4 \\ -3 & 6 \end{bmatrix} = \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

So, the inverses are:

$$A^{-1} = \begin{bmatrix} 5 & -6 \\ -4 & 5 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1/2 & -2/3 \\ -1/2 & 1 \end{bmatrix}$$

7. Given:

$$A = \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} (4\times1+1\times-3) & (4\times-1+1\times4) \\ (3\times1+1\times-3) & (3\times-1+1\times4) \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & -4+4 \\ 3-3 & -3+4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} (1\times4+-1\times3) & (1\times1+-1\times1) \\ (-3\times4+4\times3) & (-3\times1+4\times1) \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & 1-1 \\ -12+12 & -3+4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So, $BA = I_{2\times2}$.

Given:

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}, B = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix} \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_{3\times3}$$

$$BA = \begin{bmatrix} 3 & -4 & 1 \\ 2 & -4 & 1 \\ 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 2 & 3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_{3\times3}$$

Since both $AB = I$ and $BA = I$, the two matrices are inverses of eachother.

3.4 Practice Problem on solving System of Linear Equation using Matrix Method:

1.a. $\begin{cases} x + 2y = 4 \\ 3x - 5y = 1 \end{cases}$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

From $x + 2y = 4$

$x = 4 - 2y$

$3(4-2y) - 5y = 1$

$12 - 6y - 5y = 1$

$11y = 11$

$y = 1$

$x = 2$

So, $x = 2$, $y = 1$

b. $\begin{cases} 5x + y = 13 \\ 3x + 2y = 5 \end{cases}$

From, $5x + y = 13$

$y = 13 - 5x$

$3x + 2(13 - 5x) = 5$

$3x + 26 - 10x = 5$

$-7x = -21$

$x = 3$

$y = 13 - 15 = -2$

So, $x = 3, y = -2$

c. $\begin{cases} 3x + 2y = -2 \\ x + 4y = 6 \end{cases}$

From,

$x + 4y = 6$

$x = 6 - 4y$

$3(6 - 4y) + 2y = -2$

$18 - 12y + 2y = -2$

$18 - 10y = -2$

$-10y = -20$

$y = 2$

$x = 6 - 8 = -2$

So, $x = -2, y = 2$

d. $\begin{cases} 2x+3y=6 \\ x-y=1/2 \end{cases}$

From, $x-y=1/2$

$x=y+1/2$

$2\left(y+\frac{1}{2}\right)+3y=6$

$2y+1+3y=6$

$5y=5$

$y=1$

$x=1+\frac{1}{2}=\frac{3}{2}$

So, $x=3/2$, $y=1$.

e. $\begin{cases} 4x+3y=11 \\ x-3y=-1 \end{cases}$

From, add the eq^{n}s

$\cancel{4x+3y=11}$

$5x=10$

$x=2$

$2-3y=-1$

$-3y=-3$

$y=1$

So, $x=2$, $y=1$

$$\begin{cases} 2x + 3y = 8 \\ 4x + 6y = 20 \end{cases}$$

Divide second eq^n by 2:

$$2x + 3y = 10$$

We get,

$$2x + 3y = 8$$
$$2x + 3y = 10$$

Subtract,

$$0 = 2$$

$\therefore$ No solution.

4.1.1 Practice Problem: Let's Take Some Derivative:

- Problem 1: Find $f'(x)$.

1. $f(x) = x + \sqrt{x}$

$$f'(x) = \frac{d}{dx}\{x + \sqrt{x}\}$$

$$= \frac{dx}{dx} + \frac{d\sqrt{x}}{dx}$$

$$= 1 + \frac{d}{dx}\{x^{1/2}\}$$

$$= 1 + \left(\frac{1}{2}\right) x^{(1/2)-1}$$

$$= 1 + \left(\frac{1}{2}\right) x^{-1/2}$$

$$= 1 + \frac{1}{2\sqrt{x}}$$

2. $f(x) = 2 + 1/x + 1/x^2$

$$f'(x) = \frac{d(2)}{dx} + \frac{d(x^{-1})}{dx} + \frac{d(x^{-2})}{dx}$$

$$= 0 + (-1x^{-1-1}) + (-2x^{-2-1})$$

$$= -x^{-2} - 2x^{-3}$$

$$= -\frac{1}{x^2} - \frac{2}{x^3}$$

$$f(x) = \sqrt{x}\cos(x)$$

$$= \frac{d(\sqrt{x}) \cdot \cos}{d(x} \cdot \cos(x) + \sqrt{x} \cdot \frac{d(\cos(x))}{d(x)}$$

$$= \frac{d(x^{2/2})}{d(x)} \cdot \cos(x) \cdot + \sqrt{x} \cdot -\sin x$$

$$= \frac{1}{2} x^{-1/2} \cos x - \sqrt{x} \sin x.$$

$$= \frac{1}{2x^2} \cdot \cos x - \sqrt{x} \sin x.$$

$$f(x) = \ln(5x^2 + 3x)$$

$$\frac{d}{dx}[\ln(u)] = \frac{u'}{u}$$

$$u = 5x^2 + 3x.$$

$$u' = \frac{d}{dx}(5x^2 + 3x) = 10x + 3$$

$$f' = \frac{u'}{u}$$

$$= \frac{10x + 3}{5x^2 + 3x}$$

10. $f(x) = \sqrt{x^3+4x+1}$
$= (x^3+4x+1)^{1/2}$

$$\frac{d}{dx}[u^{1/2}] = \frac{1}{2}u^{-1/2} \cdot u'$$

$u = x^3+4x+1$

$u' = \frac{d}{dx}(x^3+4x+1)$

$= 3x^2+4$

$f'(x) = \frac{1}{2}(x^3+4x+1)^{-1/2} \cdot (3x^2+4)$

$$f'(x) = \frac{3x^2+4}{2\sqrt{x^3+4x+1}}$$

4. $f(x) = x^3 e^x$

$u = x^3, \; v = e^x$

$(uv)' = u'v + uv'$

$u' = \frac{d}{dx}(x^3) = 3x^2$

$v' = \frac{d}{dx}(e^x) = e^x$

$f'(x) = u'v + uv' = 3x^2e^x + x^3e^x$

$f'(x) = e^x(3x^2+x^3) = e^x(x^3+3x^2)$

$\therefore f'(x) = e^x(3x^2+x^3)$

5. $\frac{x^3}{e^x}$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}, \quad u = x^3, \; v = e^x$$

$$u' = 3x^2, \quad v' = e^x$$

$$f'(x) = \frac{3x^2 e^x - x^3 e^x}{(e^x)^2}$$

$$f'(x) = \frac{e^x(3x^2 - x^3)}{e^{2x}} = \frac{3x^2 - x^3}{e^x}$$

6. $f(x) = (3x^2 + 2x + 1)^5$

$$f'(x) = \frac{d(3x^2+2x+1)^5}{d(x)} \cdot \frac{d(3x^2+2x+1)}{d(3x^2+2x+1)}$$

$$= \frac{d(3x^2+2x+1)^5}{d(3x^2+2x+1)} \cdot \frac{d(3x^2+2x+1)}{dx}$$

$$= 5(3x^2+2x+1)^4 \cdot (6x+2).$$

7. $f(x) = (\sin x)^4$

$$\frac{d}{dx}[u^n] = nu^{n-1} \cdot u'$$

Here, $u = \sin x$, $n = 4$

$$u' = \cos x$$

$$f'(x) = 4(\sin x)^3 \cdot \cos x$$

$$f'(x) = 4\sin^3 x \cos x.$$

8. $f(x) = e^{(3x^2+2x)}$

Here,

$$u = 3x^2 + 2x$$
$$u' = 6x + 2$$
$$f'(x) = e^{3x^2+2x} \cdot (6x+2).$$
$$f'(x) = (6x+2)e^{3x^2+2x}$$

Problem 2:

1. $g(x) = x^2 - 3x + 2$
$f(x) \cdot g(x)$ at $x = 1$

$$(f.g)' = f'(x)g(x) + f(x)g'(x)$$
$$g(x) = x^2 - 3x + 2 \Rightarrow g'(x) = 2x - 3$$
$$g(1) = 1^2 - 3(1) + 2 = 0$$
$$g'(1) = 2(1) - 3$$
$$= -1$$

$$(f.g)'(1) = f'(1)g(1) + f(1)g'(1)$$
$$= (0)(0) + (5)(-1)$$
$$= -5$$

2. $\frac{f(x)}{g(x)}$ at $x = 0$

$$\left(\frac{f}{g}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

$g(0) = 0^2 - 3(0) + 2 = 2,$

$g'(0) = 2(0) - 3$

$= -3$

$$\left(\frac{f}{g}\right)'(0) = \frac{f'(0)g(0) - f(0)g'(0)}{[g(0)]^2}$$

$$= \frac{(-1)(2) - (3)(-3)}{2^2}$$

$$= \frac{-2+9}{4}$$

$$= \frac{7}{4}$$

3. $f(g(x))$ at $x = 0$.

$[f(g(x))]' = f'(g(x)) . g'(x)$

$g(0) = 2, \; g'(0) = -3$

$[f(g(x))]'(0) = f'(g(0)) . g'(0)$

$= f'(2) . (-3)$

$= 3 . (-3)$

$= -9$

4.21 Practice Problem on Maxima and Minima using derivative.

1. $g(x) = x^2 - 4x + 5$ at interval $[1, 4]$.

$$g'(x) = 2x - 4$$

To find critical points.

$$g'(x) = 0$$
$$2x - 4 = 0$$
$$x = 2$$

Testing at critical points.

$$g(2) = (2)^2 - 4 \cdot 2 + 5$$
$$= 4 - 8 + 5$$
$$= 1$$

Evaluate at Interval $[1, 4]$

$$g(1) = (1)^2 - 4 \cdot 1 + 5$$
$$= 2$$
$$g(4) = (4)^2 - 4 \cdot 4 + 5$$
$$= 16 - 16 + 5$$
$$= 5$$

min = 1 at $x = 2$

max = 5 at $x = 4$.

2. $h(x) = \frac{x^2}{2} - 2x$ at interval $[-2, 3]$.

$$h'(x) = x - 2$$

To find critical points

$$x - 2 = 0$$

$$x = 2$$

Testing at critical points

$$h(2) = \frac{4}{2} - 4$$

$$= 2 - 4$$

$$= -2$$

Evaluate at Interval $[-2, 3]$.

$$h(-2) = \frac{(-2)^2}{2} - 2(-2)$$

$$= 2 + 4$$

$$= 6$$

$$h(3) = \frac{9}{2} - 6$$

$$= 4.5 - 6$$

$$= -1.5$$

min : -2 at $x = 2$

max : 6 at $x = -2$

3. $p(x) = 3x^3 - 9x^2 + 7$ at interval $[-1, 2]$

$$p'(x) = 9x^2 - 18x$$

$$p'(x) = 9x(x-2)$$

To find critical points.

$$x = 0 \text{ or } x = 2$$

Testing at critical points.

$$p(0) = \cancel{24 - 36 + 7}\ 0 - 0 + 7$$

$$= 7$$

Evaluate at Interval $[-1, 2]$

$$p(-1) = -3 - 9 + 7$$

$$= -5$$

$$p(2) = 24 - 36 + 7$$

$$= -5$$

min: -5 at $x = -1$ or $x = 2$

max: 7 at $x = 0$

4. $f(x) = x^2 - 4x + 6$ at interval $[0, 3]$

$$f'(x) = 2x - 4$$

To find critical points.

$$2x - 4 = 0$$

$$x = 2$$

Testing at critical points.

$$f(2) = 4 - 8 + 6$$

$$= 2$$

Evaluate at Interval $[0, 3]$.

$$f(0) = 0 - 0 + 6$$

$$= 6$$

$$f(3) = 9 - 12 + 6$$

$$= 3.$$

min: 2 at $x = 2$.

max: 6 at $x = 0$.

5. $q(x) = -2x^3 + 3x^2 + 12x$ at interval $[-1, 2]$

$$q'(x) = -6x^2 + 6x + 12$$

Factor out 6:

$$q'(x) = 6(-x^2 + x + 2)$$
$$= 6(-(x-2)(x+1))$$

~~Test~~ To find critical points.

$$x = -1 \text{ or } x = 2.$$

Evaluate at Interval $[-1, 2]$

$$q(-1) = 2 + 3 - 12$$
$$= -7$$
$$q(2) = -16 + 12 + 24$$
$$= 20$$

min: -7 at $x = -1$

max: 20 at $x = 2$

4.3.3 Practice Problem on Gradients:

1. Evaluate the Gradient of a function at given points.

1. $f(x,y) = x^2 + y^2$ at:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$
$$= (2x, 2y).$$

a. $(0,0)$
$(2\times0, 2\times0) = (0,0)$

b) $(1,3)$
$(2\times1, 2\times3) = (2,6)$

c) $(-1,-5)$
$(2\times-1, 2\times-5) = (-2,-10)$

1. $f(x,y,z) = x^3z - 2y^2x + 5z$ at.

$$\frac{\partial f}{\partial x} = 3x^2z - 2y^2$$

$$\frac{\partial f}{\partial y} = -4xy$$

$$\frac{\partial f}{\partial z} = x^3 + 5$$

a) $(1, 1, -4)$

$$\frac{\partial f}{\partial x} = 3(1)^2(-4) - 2(1)^2 = -12 - 2 = -14$$

$$\frac{\partial f}{\partial y} = -4 \times 1 \times 1 = -4$$

$$\frac{\partial f}{\partial z} = 1^3 + 5 = 6$$

Gradient: $(-14, -4, 6)$

b) $(0, 1, 0)$

$$\frac{\partial f}{\partial x} = 3(0)^2(0) - 2(1)^2 = -2$$

$$\frac{\partial f}{\partial y} = -4 \times 0 \times 1 = 0$$

$$\frac{\partial f}{\partial z} = 0^3 + 5 = 5.$$

Gradient : $(-2, 0, 5)$

c) $(-3,-2,1)$

$$\frac{\partial f}{\partial x} = 3(-3)^2(1) - 2(-2)^2$$

$$= 3\times 9 - 2\times 4$$

$$= 27 - 8$$

$$= 19$$

$$\frac{\partial f}{\partial y} = -4(-3)\times(-2) = -24$$

$$\frac{\partial f}{\partial z} = (-3)^3 + 5$$

$$= -27 + 5$$

$$= -22$$

Gradient: $(19, -24, -22)$

3. $m(x,y) = \ln(x^2+y^2+1)$ at:

$$\frac{\partial m}{\partial x} = \frac{2x}{x^2+y^2+1}, \quad \frac{\partial m}{\partial y} = \frac{2y}{x^2+y^2+1}$$

a) $(0,0)$

$$\left(\frac{0}{1}, \frac{0}{1}\right) = (0,0)$$

b) $(1,-1)$

$$\left(\frac{2\times 1}{1+1+1}, \frac{2\times(-1)}{3}\right)$$

$$= \left(\frac{2}{3}, \frac{2}{3}\right)$$

c) $(-2,3)$

$$\left(\frac{2(-2)}{4+9+1}, \frac{2\times 3}{14}\right)$$

$$= \left(\frac{-4}{14}, \frac{6}{14}\right)$$

$$= \left(\frac{-2}{7}, \frac{3}{7}\right)$$

2. Problem: Using the Gradient to Evaluate Maxima and Minima of a Multivariate Function over a Specific Interval.

1. $m(x,y) = -x^2 - y^2 + 4x + 6y$ interval $x: [0, 4]$ and $y: [0, 5]$

$$\frac{dm(x,y)}{dx} = -2x + 4 \quad \text{— (i)}$$

$$\frac{dm(x,y)}{dy} = -2y + 6 \quad \text{— (ii)}$$

To find critical points.

$-2x + 4 = 0$, or $x = 2$

$-2y + 6 = 0$, or $y = 3$.

Evaluate at critical points.

$m(2,3) = -4 - 9 + 8 + 18 = 13$

$m(0,0) = 0$

$m(4,0) = -(4)^2 - (0)^2 + 4 \times 4 + 6 \times 0$

$= \cancel{-16} - 0 + \cancel{16} + 0$

$= 0$

$m(0,5) = -(0)^2 - (5)^2 + 4 \times 0 + 6 \times 5$

$= -0 - 25 + 0 + 30$

$= 5$

$m(4,5) = -(4)^2 - (5)^2 + 4 \times 4 + 6 \times 5$

$= \cancel{-16} - 25 + \cancel{16} + 30$

$= 5$

Max 13 at $(2,3)$ Min 0 at $(0,0)$ & $(4,0)$

2. $g(x,y) = xy + x^2 - 4y + 6$ interval $x: [-2, 2]$ and $y: [1, 3]$

$$\frac{dg(x,y)}{dx} = y + 2x$$

$$\frac{dg(x,y)}{dy} = x - 4$$

To find critical points.

$y + 2x = 0$, $x - 4 = 0$ (outside interval discord).

$y = -2x$ $\quad x = 4$

No critical point in the interval.

Check boundaries and corners:

$g(-2, 1) = -2 + 4 - 4 + 6 = 4$

$g(-2, 3) = -6 + 4 - 12 + 6 = -8$

$g(2, 1) = 2 + 4 - 4 + 6 = 8$

$g(2, 3) = 6 + 4 - 12 + 6 = 4$

$\therefore$ Max: $(2, 1)$, Min: $(-2, 3)$

3. $p(x,y) = x^3 - 3xy + 4$ interval $x: [0,2]$ and $y: [0,3]$

$$\frac{\partial p}{\partial x} = 3x^2 - 3y, \quad \frac{\partial p}{\partial y} = -3x.$$

Set, gradient $= 0$;

$-3x = 0 \Rightarrow x = 0$

$3x^2 - 3y = 0 \Rightarrow y = x^2$

Critical points inside interval : $x = 0, y = 0$.

Now,

Evaluating function at critical point and corners:

Corners: $(0,0), (0,3), (2,0), (2,3)$.

$p(0,0) = 0 - 0 + 4 = 4$

$p(0,3) = 0 - 0 + 4 = 4$

$p(2,0) = 8 - 0 + 4 = 12$

$p(2,3) = 8 - 18 + 4 = -6$

~~∴ Mat~~ ∴ Max 12 at $(2,0)$, Min -6 at $(2,3)$

4. $h(x,y) = 2x^2 + 3y^2 - 12x - 18y + 20$ interval $x: [1, 5]$ and $y: [0, 4]$.

$$\frac{\partial h}{\partial x} = 4x - 12 \quad , \quad \frac{\partial h}{\partial y} = 6y - 18$$

Set, gradient $= 0 \Rightarrow$ critical point

$$4x - 12 = 0 \Rightarrow x = 3$$
$$6y - 18 = 0 \Rightarrow y = 3$$

Evaluating function at critical point & corners.

$h(3,3) = 29 + 39 - 36 - 54 + 20 = -25$ (Min).

Check corners:

$(1, 0) = 2 - 12 + 0 - 0 + 20 = 10$ (Max)

$(1, 4) = 2 - 12 + 48 - 72 + 20 = -14$

$(5, 0) = 50 - 60 + 0 - 0 + 20 = 10$

$(5, 4) = 50 - 60 + 48 - 72 + 20 = -14$.

$\therefore$ Max 10 at $(1, 0)$ and Min -25 at $(3, 3)$.

5. $f(x,y) = x^2 + y^2 - 4x - 6y + 13$ interval :
$x : [1, 3]$ and $y : [1, 4]$.

$$\frac{\partial f}{\partial x} = 2x - 4, \quad \frac{\partial f}{\partial y} = 2y - 6$$

set, gradient $= 0 \rightarrow$ critical point:

$$2x - 4 = 0 \Rightarrow x = 2$$
$$2y - 6 = 0 \Rightarrow y = 3$$

Evaluating function at critical point & corner.

$f(2,3) = 4 + 9 - 8 - 18 + 13 = 0$ (Min)

Corners: $(1,1) = 1 + 1 - 4 - 6 + 13 = 5$ (Max)

$(3,1) = 9 + 1 - 12 - 6 + 13 = 5$ (Max)

$(1,4) = 1 + 16 - 4 - 24 + 13 = 2$

$(3,4) = 9 + 16 - 12 - 24 + 13 = 2$

$\therefore$ Max 5 at $(1,1)$, $(3,1)$, Min 0 at $(2,3)$.